

UIT-RGPV (Autonomous) Bhopal

Subject code-CS-601

Subject: Principles of Programming Languages

Semester: VI

For credits & marks refer your scheme

Course Objective-The purpose of this subject is to cover the underlying concepts and techniques used in Programming Languages. It provides general idea related to operating & Programming environment.

CO1	Understand different Programming Paradigms and basic concepts for the Implementation of Programming Language.
CO2	Understand data types and concept of binding, type checking and conditional statements used in programming languages.
CO3	Understand Subprograms, Blocks used in programming languages and Design issues of subprograms.
CO4	Understand the basic concepts of Abstract Data types and object oriented programming Languages.
CO5	Understand the concepts of Exception handling , Logic Programming & Functional Programming

COURSE CONTENT:

UNIT-I

Language Evaluation Criteria, influences on Language design, Language categories, Programming Paradigms – Imperative, Object Oriented, functional Programming, Logic Programming. Programming Language Implementation – Compilation and Virtual Machines, programming environments. Issues in Language Translation: Syntax, Semantics, Stages, analysis and synthesis, Parse Tree, CFG and BNF grammar.

UNIT-II

Data types: Introduction, primitive, character, user defined, array, associative, record, union, pointer and reference types, design and implementation uses related to these types. Names, Variable, concept of binding, type checking, strong typing, type compatibility, named constants, variable initialization. Sequence control with Expressions, Conditional Statements, Loops, Exception handling.

UNIT-III

Subprograms and Blocks: Fundamentals of sub-programs, Scope and lifetime of variable, static and dynamic scope, Design issues of subprograms and operations, local referencing environments, parameter passing methods, overloaded sub-programs, generic sub-programs, design issues for functions overloaded operators, co routines.

UNIT-IV

Abstract Data types: Abstractions and encapsulation, introductions to data abstraction, Static and Stack- Based Storage management. heap based storage management. Garbage Collection. object oriented programming in small talk, C++, Java, C#, PHP, Perl . Concurrency: Subprogram level concurrency, semaphores, monitors, message passing, Java threads, C# threads.

UNIT – V

Exception handling, Exceptions, exception Propagation, Exception handler in C++ and Java.

Logic Programming Language: Introduction and overview of logic programming, basic elements of prolog, application of logic programming. Functional Programming Languages: Introduction, fundamentals. Introduction to 4GL.

References:

1. Sebesta, "Concept of programming Language", Pearson Edu.
2. Louden, "Programming Languages: Principles & Practices", Cengage Learning
3. Tucker, "Programming Languages: Principles and paradigms", Tata McGraw –Hill
4. Terrance W Pratt, "Programming Languages: Design and Implementation" Pearson Edu.
5. Cavlo Ghezzi & Mehdi Jazayeri "Programming Languages Concepts", Willey India
6. E Horowitz, "Programming Languages", 2nd Edition, Addison Wesley

UIT-RGPV (Autonomous) Bhopal

Subjectcode-CS-602
Semester:VI

Subject: Computer Networking

For credits & marks refer your scheme

Course Outcomes: At the end of the course student will be able to

CO-1	Describe different aspects of network, protocols and network design models. Analyze computer network protocol hierarchy of OSI/TCP Models.
CO-2	Examine various Data Link Layer Design Issues and Data Link protocols. Determine and simulate various Data Link Layer protocols to avoid collision and congestion.
CO-3	Analyze MAC Sublayer and LAN Technologies. Elaborate and interpret various Distributed Random Access Schemes/Contention Schemes for Data Services and LANs. Analyze and compare different LAN protocols.
CO-4	Determine various Network Layer Design Issues. Implement routing and congestion control algorithms. Compare and select appropriate routing algorithm for a network. Differentiate IPv4 and IPv6 addressing schemes, Knowledge of SDN Architecture.
CO-5	Examine the important aspects and functions of Transport Layer, Presentation layer and Application Layer in internetworking.

COURSE CONTENT:

Unit –I

Computer Network: Definitions, goals, components, Architecture, Classifications & Types. Layered Architecture: Protocol hierarchy, Design Issues, Interfaces and Services, Connection Oriented & Connectionless Services, Service primitives, Design issues & its functionality. ISOOSI Reference Model: Principle, Model, Descriptions of various layers and its comparison with TCP/IP. Network standardization. Queueing Models: Little's Theorem, Queueing System: M/M/1, M/M/m, M/M/∞, M/M/m/m, M/G/1

Unit-II

Data Link Layer: Need, Services Provided, Framing, Flow Control, Error control. Data Link Layer Protocol: Elementary & Sliding Window protocol: 1-bit, Go-Back-N, Selective Repeat, Hybrid ARQ. Bit oriented protocols: SDLC, HDLC, BISYNC, LAP and LAPB. Protocol verification: Finite State Machine Models & Petri net models.

Unit-III

MAC Sublayer: MAC Addressing, Binary Exponential Back-off (BEB) Algorithm, Distributed Random Access Schemes/Contention Schemes: for Data Services (ALOHA and Slotted- ALOHA), for Local-Area Networks (CSMA, CSMA/CD, CSMA/CA), Collision Free Protocols: Basic Bit Map, BRAP, Binary Count Down, MLMA Limited Contention Protocols: Adaptive Tree Walk, URN Protocol, High Speed LAN: Fast Ethernet, Gigabit Ethernet, FDDI, Performance Measuring Metrics. IEEE Standards 802 series & their variant.

Unit-IV

Network Layer: Need, Services Provided, Design issues, Routing algorithms: Least Cost Routing algorithm, Dijkstra's algorithm, Bellman-ford algorithm, Hierarchical Routing, Broadcast Routing, Multicast Routing, Congestion Control Algorithms: General Principles of Congestion control, Prevention Policies, Congestion Control in Virtual-Circuit Subnets, Congestion Control in Datagram subnets. IP protocol, IP Addresses, Comparative study of IPv4 & IPv6, Mobile IP.

Software-Defined Networking (SDN): SDN Architecture, Control Plane, Data Plane, Services (As a Controller/Switches), SDN Controller, Traditional Network Vs. SDN, Interfacing Between Control & Data Planes, OpenFlow, SDN Message Interfacing, Packet Processing, N/W in a Computer Vs. N/W with Controller in a Computer.

Unit-V

Transport Layer: Design Issues, UDP Header Format, Per-Segment Checksum, Carrying Unicast/Multicast Real-Time Traffic, TCP: Connection Management, Reliability of Data Transfers, TCP Flow Control, TCP Congestion Control, TCP Header Format, TCP Timer Management. Session layer: Authentication, Authorization, Session layer protocol (PAP, SCP, H.245). Presentation layer: Data conversion, Character code translation, Compression, Encryption and Decryption, Presentation layer protocol (LPP, Telnet, X.25 packet Assembler/Disassembler). Application Layer: WWW and HTTP, FTP, SSH, Email (SMTP, MIME, IMAP), DNS, Network Management(SNMP).

List of Experiments:

1. Study of Different Type of LAN & Network Equipments.
2. Study and Verification of standard Network topologies i.e. Star, Bus, Ring etc.
3. LAN installations and Configurations.
4. Write a program to implement various types of error correcting techniques.
5. Write a program to Implement various types of framing methods.
6. Study of Tool Command Language (TCL).
7. Study and Installation of Standard Network Simulator: N. S-2, N. S-3, OpNet, QualNet etc.
8. Study & Installation of ONE (Opportunistic Network Environment) Simulator for High Mobility Networks.
9. Configure 802.11 WLAN.
10. Implement & Simulate various types of routing algorithm.
11. Study & Simulation of MAC Protocols like Aloha, CSMA, CSMA/CD and CSMA/CA using Standard Network Simulators.
12. Study of Application layer protocols- DNS, HTTP, HTTPS, FTP and TelNet.
13. Study & Installation of Wireshark
14. Socket Programming

References:

1. Andrew S. Tanenbaum, David J. Wetherall, "Computer Networks" Pearson Education.
2. Dimitri Bertsekas, Robert Gallager, "Data Networks", PHI Publication, Second Edition.
3. Kaveh Pahlavan, Prashant Krishnamurthy, "Networking Fundamentals", Wiley Publication.
4. Uyless Black, "Computer Networks", PHI Publication, Second Edition.
5. Ying-Dar Lin, Ren-Hung Hwang, Fred Baker, "Computer Networks: An Open Source Approach", McGraw Hill.
6. <https://www.wireshark.org>
7. Hands-On Network Programming with C, www.packed.com

UIT-RGPV (Autonomous) Bhopal

Subject code-CS-613(B)

Subject: Digital Image Processing

Semester: VI

For credits & marks refer your scheme

CO 1	Describe in-depth about theories, models, and algorithms in Digital Image Processing
CO 2	Compare and contrast different Image Transformations techniques.
CO 3	Examine the consequences of Image enhancement in spatial domain and frequency domain.
CO 4	Explain different types of Image Compression Algorithms.
CO 5	Describes advanced Morphology and Segmentation Algorithms.

COURSE CONTENT:

Unit – I

Introduction

Image Model, Image Sensing and Acquisition, Sampling and Quantization, Digital image representation, Mathematical Tool for Digital Image Processing, Image File Formats, Colour Fundamentals and Models, Basic Intensity Transformation Functions

Unit – II

Image Transformations and Filtering

Fundamentals of Spatial Filtering, Smoothing and Sharpening Spatial Filters. Sampling and Fourier Transform, 2 D- Discrete Fourier Transform (DFT), DCT, Hadamard and Walsh transforms. Filtering in the Frequency Domain, Smoothing and Sharpening Frequency Domain Filters

Unit – III

Image Enhancement and Image Restoration

Image enhancement in spatial domain and frequency domain. Histogram Processing, Histogram equalization. Low pass filtering and High pass filtering. Image Degradation/ Restoration Process, Image restoration techniques, Noise Models, Image restoration using Spatial Filtering and Frequency Domain Filtering, Estimating the Degradation Function

Unit – IV

Image Compression

Image Compression basics, Types of Image Compression Algorithms, Lossless Compression Algorithms, Lossy Compression Algorithms, variable length coding, Huffman coding, LZW, Image and Video Compression Standards, JPEG, MPEG.

Unit – V

Morphology and Segmentation

Erosion and Dilation, Opening and Closing, Morphological Algorithms, Grey Scale Morphology, Point, Line and Edge segmentation, Edge linking and boundary detection, Hough transform, Thresholding, Region based Segmentation, Segmentation using Morphological Watersheds.

Books:

1. Rafael C. Gonzales and Richard E. Woods, “Digital Image Processing”
2. Anil K. Jain, “Fundamentals of Digital Image Processing”

UIT-RGPV (Autonomous) Bhopal

Subject code-613(D)

Subject: Object Oriented Analysis and Design

Semester: VI

For credits & marks refer your scheme

Course Objective-

Students able to

CO1	Define the fundamentals of OO approach.
CO2	Illustrate about domain models and conceptual classes
CO3	Select an appropriate design pattern
CO4	Explain Translation Object Oriented design into implementation
CO5	Construct projects using UML diagrams

COURSE CONTENT:

Unit I:

Overview of Object-Oriented concepts: Objects and classes, abstraction, generalization and inheritance, encapsulation, aggregation, abstract classes, polymorphism, link, and association, Need for object-oriented approach.

Unit II:

System design life cycle, object-oriented S/W development process model, Object Oriented Analysis, Object Modelling Technique (OMT): object model, dynamic model, functional model, relationship among models, object diagrams, state diagrams, data flow diagrams.

Unit III:

Object oriented Design: Overview of object design, Combination the models, designing algorithms, design optimization, Implementation of control, Adjustment, Design of association, object representation, physical packaging, documenting design decision, comparison of use-case driven approach.

Unit IV:

Translation Object Oriented design into implementation, Programming style, documentation, characterization of object-oriented languages, Comparison of object-oriented language like C++, JAVA. Distributed object model: CORBA and COM / DCOM

Unit V:

Unified Modelling Language (UML): Package Diagrams, Component Diagrams, Deployment Diagrams, Use Case Diagrams, Activity Diagrams, Class Diagrams, Sequence Diagrams, Interaction Overview Diagrams, State Machine Diagrams, Timing Diagrams, Object Diagrams, Communication Diagrams

Object oriented Database: Relational vs. object-oriented database, the architecture of object-oriented database, query language for Object Oriented database.

References

1. Bertrand Meyer, Object Oriented Software Construction, Prentice-Hall.
2. Grady Booch, Object Oriented Analysis and Design, Addison-Wesley.
3. Grady Booch, James Rumbaugh and Ivar Jacobson, Unified Modeling Language Guide, Addison-Wesley.
4. Saba Zamir, Handbook of Object Technology, CRC Press.
5. Satzinger, Jackson and Burd, "Object oriented Analysis and design with the Unified Process", CENGAGE Learning

UIT-RGPV (Autonomous) Bhopal

Subject code-CS-614(A)

Subject: Advance Computer Architecture

Semester: VI

For credits & marks refer your scheme

Course Outcomes: At the end of the course student will be able to:

CO 1	Classify various system architecture and interconnection networks.
CO 2	Understand the concept of Instruction set architecture, CISC Scalar Processors , RISC Scalar Processors, Memory Hierarchy, Coherence and Locality, Memory capacity planning.
CO 3	Understand the concept of various pipeline processor with branch handling techniques
CO 4	Understand the concept of Directory based protocols, Vector Processing Principles, and Principles of Multithreading.
CO 5	Illustrate various Parallel Programming Models and Software tools.

COURSE CONTENT:

Unit-I

Flynn's Classification, System Attributes to Performance, Parallel computer models -Multiprocessors and multicomputer, Multifactor and SIMD Computers. Data and resource dependences, Hardware and software parallelism, Program partitioning and scheduling, Grain size and latency, Control flow, data flow and Demand driven mechanisms. Static interconnection networks, Dynamic interconnection Networks: Bus Systems, Crossbar Switch, Multiport Memory, Multistage and Combining Networks

Unit-II

Instruction set architecture, CISC Scalar Processors , RISC Scalar Processors, VLIW architecture, Memory Hierarchy, Inclusion, Coherence and Locality, Memory capacity planning.

Interleaved memory organization- memory interleaving, pipelined memory access, Bandwidth and Fault Tolerance. Backplane Bus System: Backplane bus specification, Addressing and timing protocols, Arbitration transaction and interrupt.

Unit-III

Linear pipeline processor, Nonlinear pipeline processor, Instruction pipeline design, Mechanisms for instruction pipelining, pipeline hazards, Dynamic instruction scheduling - score boarding and Tomosulo's algorithm, Branch handling techniques, Arithmetic Pipeline Design, Static arithmetic pipeline, Multifunctional arithmetic pipelines. Superscalar pipeline design, Super pipeline processor design.

Unit-IV

Cache coherence, Snoopy protocols, Directory based protocols. Message routing schemes in multicomputer network, deadlock and virtual channel. Vector Processing Principles, Vector instruction types, Vector-access memory schemes. Vector supercomputer architecture, SIMD organization: distributed memory model and shared memory model. Principles of Multithreading: Multithreading Issues and Solutions, Multiple-Context Processors

Unit-V

Parallel Programming Models, Shared-Variable Model, Message-Passing Model, Data-Parallel Model, Object-Oriented Model, Functional and Logic Models, Parallel Languages and Compilers, Language Features for Parallelism, Parallel Programming Environment, Software Tools and Environments.

Suggested Reading:

1. Kai Hwang, “Advanced computer architecture”, TMH.
2. J.P.Hayes, “computer Architecture and organization”; MGH.
3. V.Rajaraman & C.S.R.Murthy, “Parallel computer”; PHI Learning.
4. Kain,”Advance Computer Architecture: - A System Design Approach”, PHI Learning
5. M.J Flynn, “Computer Architecture, Pipelined and Parallel Processor Design”; Narosa Publishing.
6. Hwang and Briggs, “Computer Architecture and Parallel Processing”; MGH.
7. David E. Callav & Jaswinder Pal Singh Marge Kaufmann”Advance Computer Architecture”.
8. Sajjan G. Shiva, Taylar & Francis, “Advance Computer Architecture

UIT-RGPV (Autonomous) Bhopal

Subject code-CS-614(B)

Subject: Mobile Application Development

Semester: VI

For credits & marks refer your scheme

Course Outcomes: At the end of the course student will be able to:

CO 1	Demonstrate knowledge of different issues in mobile communication and apply this knowledge to optimized mobile communication network.
CO 2	Describe the design and architecture of a mobile application and Identify the challenges that mobile programming has in providing an effective user interface.
CO 3	Understand use of mobile agents and peer to peer architecture to meet the requirements of a mobile application.
CO 4	Understand the concepts of Synchronization and Replication of Mobile Data.
CO 5	Address the limitations and challenges of working in a mobile environment and to test the mobile application in different environments.

COURSE CONTENT:

Unit-I

Mobile Communication Fundamentals

Introduction, issues in mobile communications, Wireless telephony: cellular concept, GSM: air-interface, channel structure, location management: HLR-VLR, handoffs, channel allocation in cellular systems, CDMA, GPRS, Wireless LAN Overview: MAC issues, IEEE 802.11, Blue Tooth, Wireless multiple access protocols, TCP over wireless, Wireless applications, Mobile IP, WAP: Architecture, protocol stack, applications.

Unit-II

Mobile Applications Development Frameworks and Tools

Introduction of Mobile Applications, Types and Benefits of a Mobile App, Mobile Platforms, deployment on Apple iOS with versions, Android, Windows phone application using development platforms: Worklight, kendo, Appcon, Xcode, Xpage, Architecture of Mobile Software Applications, N-Tier Client–Server Frameworks and Tools, Java, BREW, Windows CE, WAP, Symbian EPOC, Publishing Frameworks, Mobile User Interface Design, Building Generic User Interfaces, mobile apps in the cloud.

Unit-III

Mobile Agents and Peer-to-Peer Architectures for Mobile Applications

Mobile Agents for Mobile Computing, Applications of Mobile Agents to Mobile Applications and Implementation Tools, Techniques for Agent-Based Software, Peer-to-Peer Applications for Mobile Computing, security and fault tolerance.

Unit-IV

Synchronization and Replication of Mobile Data

Taxonomy of Replication and Synchronization, Data Replication and Synchronization for Mobile Applications, SyncML, WebDAV, Mobile Agents, Replication, and Synchronization, Location Information Modeling, Problems with Building Location-Based Applications, Utilizing Location-Based Services with Mobile Applications, UML-Based Development Cycle for Mobile Applications,

Unit-V

Testing Mobile Applications, Validating the Mobile Use Cases before Development, The Effect of the Dimensions of Mobility on Software Testing, Stress Testing and Scalability Issues, Testing Location-Based Functionality, Android as your mobile platform, installation, Configuring of Eclipse and the Android SDK, Additional SDK Components, application layout and Android app development, Android user interface elements, Android Virtual Device, Connection to Google play.

References :

1. Reza b'far, Mobile computing Principles Designing and developing Mobile applications with Uml and xml, Cambridge university press.
2. Jeff Mcwherter, Scott Gowell, Professional Mobile application development, Wrox, John Wiley & Sons, Inc..
3. Richard Rodger, Beginning mobile application Development in the cloud, John Wiley & Sons, Inc.
4. J. Schiller, Mobile Communication, Addison Wesley.

UIT-RGPV (Autonomous) Bhopal

Subject code-CS-614(C)

Subject: Deep & Reinforcement Learning

Semester: VI

For credits & marks refer your scheme

After completing the course student should be able to

CO 1	Describe in-depth about theories, models and algorithms in machine learning.
CO 2	Compare and contrast different learning algorithms with parameters.
CO 3	Examine the nature of a problem at hand and find the appropriate learning algorithms and it's parameters that can solve it efficiently enough
CO 4	Design and implement of deep and reinforcement learning approaches for solving real-life problems.
CO 5	Describes advanced Q learning algorithms and reinforcement learning .

COURSE CONTENT:

Unit 1:

History of Deep Learning, McCulloch Pitts Neuron, Thresholding Logic, Activation functions, Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMSProp, Adam, Eigenvalue Decomposition. Recurrent Neural Networks, Backpropagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, GRU, LSTMs, Encoder Decoder Models, Attention Mechanism, Attention overimages.

Unit 2:

Autoencoders and relation to PCA, Regularization in autoencoders, Denoising autoencoders, Sparse autoencoders, Contractive autoencoders, Regularization: Bias Variance Tradeoff, L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods, Dropout, Batch Normalization, Instance Normalization, Group Normalization.

Unit 3:

Greedy Layerwise Pre-training, Better activation functions, Better weight initialization methods, Learning Vectorial Representations Of Words, Convolutional Neural Networks, LeNet, AlexNet, ZF-Net, VGGNet, GoogLeNet, ResNet, Visualizing Convolutional Neural Networks, Guided Backpropagation, Deep Dream, Deep Art, Recent Trends in Deep Learning Architectures.

Unit 4:

Introduction to reinforcement learning(RL), Bandit algorithms – UCB, PAC, Median Elimination, Policy Gradient, Full RL & MDPs, Bellman Optimality, Dynamic Programming - Value iteration, Policy iteration, and Q-learning & Temporal Difference Methods, Temporal-Difference Learning, Eligibility Traces, Function Approximation, Least Squares Methods

Unit 5:

Fitted Q, Deep Q-Learning , Advanced Q-learning algorithms , Learning policies by imitating optimal controllers , DQN & Policy Gradient, Policy Gradient Algorithms for Full RL, Hierarchical RL,POMDPs, Actor-Critic Method, Inverse reinforcement learning, Maximum Entropy Deep Inverse Reinforcement Learning, Generative Adversarial Imitation Learning,Recent Trends in RL Architectures

Text Books:

1. Deep Learning, An MIT Press book, Ian Goodfellow and YoshuaBengio and Aaron Courville
2. Pattern Classification- Richard O. Duda, Peter E. Hart, David G. Stork, John Wiley & Sons Inc.
3. Reinforcement Learning: An Introduction, Sutton and Barto, 2nd Edition.
4. Reinforcement Learning: State-of-the-Art, Marco Wiering and Martijn van Otterlo, Eds

UIT-RGPV (Autonomous) Bhopal

Subject code-CS-615

Subject: Project Management

Semester: VI

For credits & marks refer your scheme

Students able to

CO1	Understand project characteristics and various stages of a project and the conceptual clarity about project organization and feasibility analyses
CO2	Analyse the learning and understand techniques for Project planning, scheduling and Execution Control.
CO3	Understand efficient techniques for Project Monitoring and control.
CO4	Explain various testing techniques to the Software to ensure Software quality assurance
CO5	Create the risk management plan and analyze the role of stakeholders

COURSE CONTENT:

Unit-I

Introduction and Software Project Planning: Fundamentals of Software Project Management (SPM), Need Identification, Vision and Scope document, Project Management Cycle, SPM Objectives, Management Spectrum, SPM Framework, Software Project Planning, Planning Objectives, Project Plan, Types of project plan, Structure of a Software Project Management Plan, Software project estimation, Estimation methods, Estimation models, Decision process.

Unit-II

Project Organization and Scheduling Project Elements: Work Breakdown Structure (WBS), Types of WBS, Functions, Activities and Tasks, Project Life Cycle and Product Life Cycle, Ways to Organize Personnel, Project schedule, Scheduling Objectives, Building the project schedule, Scheduling terminology and techniques, Network Diagrams: PERT, CPM, Bar Charts: Milestone Charts, Gantt Charts. (SPI).

Unit-III

Project Monitoring and Control: Dimensions of Project Monitoring & Control, Earned Value Analysis, Earned Value Indicators: Budgeted Cost for Work Scheduled (BCWS), Cost Variance (CV), Schedule Variance (SV), Cost Performance Index (CPI), Schedule Performance Index (SPI), Interpretation of Earned Value Indicators, Error Tracking, Software Reviews, Types of Review: Inspections, Deskchecks, Walkthroughs, Code Reviews, Pair Programming.

Unit-IV

Software Quality Assurance and Testing Objectives: Testing Principles, Test Plans, Test Cases, Types of Testing, Levels of Testing, Test Strategies, Program Correctness, Program Verification & validation, Testing Automation & Testing Tools, Concept of Software Quality, Software Quality Attributes, Software Quality Metrics and Indicators, The SEI Capability Maturity Model CMM), SQA Activities, Formal SQA Approaches: Proof of correctness, Statistical quality assurance.

Unit-V

Project Management and Project Management Tools Software Configuration Management: Software Configuration Items and tasks, Baselines, Plan for Change, Change Control,

Change Requests Management, Version Control, Risk Management: Risks and risk types, Risk Breakdown Structure (RBS), Risk Management Process: Risk identification, Risk analysis, Risk planning, Risk monitoring, Cost Benefit Analysis, Software Project Management Tools: CASE Tools, Planning and Scheduling Tools, MS-Project.

References

1. M. Cotterell, Software Project Management, Tata McGraw-Hill Publication.
2. Royce, Software Project Management, Pearson Education
3. Kieron Conway, Software Project Management, Dreamtech Press
4. S. A. Kelkar, Software Project Management, PHI Publication.
5. Harold R. Kerzner, Project Management "A Systems Approach to Planning, Scheduling, and Controlling" Wiley.
6. Mohapatra, Software Project Management, Cengage Learning.
7. P.K. Agarwal, SAM R., Software Project Management, Khanna Publishing House